Poisson Distribution

Note that,

$$e^{\lambda} = \sum_{x=0}^{\infty} \frac{\lambda^x}{x!}$$

$$\Rightarrow \sum_{x=0}^{\infty} e^{-\lambda}\frac{\lambda^x}{x!} = e^{-\lambda}e^{\lambda} = 1 \quad \text{, for } \lambda > 0.$$

$$\therefore e^{-\lambda}\frac{\lambda^x}{x!} > 0 \quad \forall x = 0, 1, 2, \ldots$$

The terms of the series $\left\{\frac{e^{-\lambda}\lambda^x}{x!}\right\}, x = 0, 1, \ldots$ forms a PMF. The PMF is known as Poisson Distribution.

1 Definition

A counting random variable X is said to have a Poisson Distribution if its PMF is given by,

$$f(x) = \begin{cases} \frac{e^{-\lambda}\lambda^x}{x!} & , x = 0, 1, \ldots, \lambda > 0 \\ 0 & \text{, otherwise} \end{cases}$$

Remark

For $X = 0, f(0) = e^{-\lambda}\frac{\lambda^0}{0!} = 1$

$$P(X = 0) = 1$$

$\therefore X$ is degenerated at 0 .

2 Moments

First, we will focus on factorial moments. $[x! = (x)_r(x-r)!]$

$$\begin{aligned} E\left(X_r\right) &= \sum_{x=0}^{\infty}(x)_r e^{-\lambda}\frac{\lambda^x}{x!} \\ &= \sum_{x=r}^{\infty} e^{-\lambda}\frac{\lambda^x}{(x-r)!} \\ &= \lambda^r \sum_{y=x-r=0}^{\infty} \frac{e^{-\lambda}\lambda^y}{y} \\ &= \lambda^r e^{-\lambda}e^{\lambda} \\ &= \lambda^r \end{aligned}$$

$$E(X) = E\left(X_1\right) = \lambda$$

$$\begin{aligned}\text{Var}(X) &= E\left(X^2\right) - E^2(X) \\ &= E(X(X-1)+X) - \lambda^2 \\ &= E\left(X_2\right) + E(X) - \lambda^2 \\ &= \lambda^2 + \lambda - \lambda^2 = \lambda\end{aligned}$$

Note

$$\begin{aligned}&\text{For } \text{Bin}(n,p) \to E(x) > \text{var}(x) \\ &\text{For } \text{NB}(n,p) \to E(x) < \text{var}(x) \\ &\text{For } \text{Poi}(\lambda) \to E(x) = \text{var}(x)\end{aligned}$$

3 Mode

$$\begin{aligned}\frac{f(x)}{f(x-1)} &= \frac{e^{-\lambda}\lambda^x}{x!}\frac{(x-1)!}{e^{-\lambda}\lambda^{x-1}} \\ &= \frac{\lambda}{x} \gtreqless 1 \text{ according as } \lambda \gtreqless cx.\end{aligned}$$

Case I: λ is an integer. Let $\lambda = k$.

$$\begin{aligned}&\frac{f(1)}{f(0)} > 1 \Rightarrow f(1) > f(0) \\ &\frac{f(2)}{f(1)} > 1 \Rightarrow f(2) > f(1) \\ &\vdots \\ &\frac{f(x)}{f(x-1)} = 1 \Rightarrow f(x) = f(x-1) \\ &\frac{f(x+1)}{f(x)} < 1 \Rightarrow f(x+1) < f(x)\end{aligned}$$

So, $f(0) < f(1) < \ldots < f(x-1) = f(x) > f(x+1) > f(x+2) > \ldots$

Mode is at λ and $\lambda - 1$.

Case II: λ is not an integer, let $k = [\lambda]$.

$$f(0) < f(1) < \ldots < f(k-1) < f(k) > f(k+1) > \ldots$$
$$\therefore k = [\lambda] \text{ is the mode.}$$

4 Mean Deviation about Mean

$$\begin{aligned}\text{MD}_\lambda = E|X-\lambda| &= \sum_{x=0}^{\infty} |x-\lambda| \frac{e^{-\lambda}\lambda^x}{x!} \\ &= \sum_{x=0}^{[\lambda]} (\lambda - x)\frac{e^{-\lambda}\lambda^x}{x!} + \sum_{x=[\lambda]+1}^{\infty} (x-\lambda)\frac{e^{-\lambda}\lambda^x}{x!}, \quad \text{as } K = [\lambda] \text{ is the mode.} \\ &= 2e^{-\lambda} \sum_{x=k+1}^{\infty} (x-\lambda)\frac{\lambda^x}{x!} \\ &= 2e^{-\lambda} \sum_{x=k+1}^{\infty} \left\{ \frac{\lambda x}{(x-1)!} - \frac{\lambda^{x+1}}{x!} \right\}\end{aligned}$$

$$
\begin{aligned}
&= 2e^{-\lambda} \sum_{x=k+1}^{\infty} (v_x - v_{x+1}), \quad v_x = \frac{\lambda x}{(x-1)!} \\
&= 2e^{-\lambda} (v_{k+1} - v_{k+2} + v_{k+2} - v_{k+3} + \cdots) \\
&= 2e^{-\lambda} \frac{\lambda^{k+1}}{k!} \\
&= 2e^{-\lambda} \frac{\lambda \lambda^k}{k!} = 2\lambda e^{-\lambda} \frac{\lambda^{[\lambda]}}{[\lambda]!}
\end{aligned}
$$

Remark

$$\frac{\text{MD}_\lambda}{\text{SD}} = \frac{2\lambda e^{-\lambda} \lambda^{[\lambda]}}{[\lambda]!} \frac{1}{\sqrt{\lambda}}$$

If $\lambda = 1$, $\quad \dfrac{\text{MD}}{\text{SD}} = \dfrac{2}{e} = 2e^{-1}$.

5 Relationship with Compound Binomial Distribution

Suppose, the probability of an insect laying n eggs is $\dfrac{e^{-\lambda}\lambda^n}{n!}$ and the probability that the egg developing is p. Assuming natural independence of the eggs, show that the probability of the total survivor is given by $\text{Poi}(\lambda p)$.

Let, N be the no of eggs laid down by an insect.

$$N \sim \text{Poi}(\lambda).$$

Let, X be the number of survivors,

$$
\begin{aligned}
P(X = x \mid N = n) &= \begin{cases} \binom{n}{x} p^x q^{n-x} & x = 0, \ldots n \\ 0 & \text{ow} \end{cases} \\
P(X = x) &= \textstyle\sum_{n=0}^{\infty} P(X = x \mid N = n) P(N = n) \quad \begin{cases} n = 0, 1, \ldots \infty \\ x = 0, 1, \cdots, n \end{cases} \\
&= \sum_{n=x}^{\infty} \binom{n}{x} p^x q^{n-x} e^{-\lambda} \frac{\lambda^n}{n!} \begin{cases} x < n < \infty; 0 < n < \infty; \\ 0 < x < n < \infty \end{cases} \\
&= \frac{e^{-\lambda} \lambda^x p^x}{x!} \sum_{n=x}^{\infty} \frac{\lambda^{n-x} q^{n-x}}{(n-x)!} \\
&= \frac{e^{-\lambda} (\lambda p)^x}{x!} \sum_{n-x=0}^{\infty} \frac{(\lambda q)^{n-x}}{(n-x)!} \\
&= e^{-\lambda} \frac{(\lambda p)^x}{x!} e^{\lambda(1-p)} \\
&= \frac{e^{-\lambda p} (\lambda p)^x}{x!}, x = 0, 1, \ldots \\
X &\sim \text{Poi}(\lambda p).
\end{aligned}
$$

1. Find the value of $E\left(\dfrac{1}{1+X}\right)$.

$$
\begin{aligned}
E\left(\frac{1}{1+X}\right) &= \sum_{x=0}^{\infty} \frac{1}{1+x} e^{-\lambda} \frac{\lambda^x}{x!} \\
&= \sum_{x=0}^{\infty} \frac{e^{-\lambda}}{\lambda} \frac{\lambda^{(x+1)}}{(x+1)!} = \frac{e^{-\lambda}}{\lambda} \sum_{x+1=y=1}^{\infty} \frac{\lambda^y}{y!}
\end{aligned}
$$

$$= \frac{e^{-\lambda}}{\lambda}\left(1 - e^{-\lambda}\right)$$

6 Recursion Relation of Poisson Distribution

$$\begin{aligned}
\mu_r &= E(X-\lambda)^r = \sum_{x=0}^{\infty}(x-\lambda)^r e^{-\lambda}\frac{\lambda^x}{x!} \\
\frac{\partial \mu_r}{\partial \lambda} &= -\sum_{x=0}^{\infty} r(x-\lambda)^{r-1}e^{-\lambda}\frac{\lambda^x}{x!} - \sum_{x=0}^{\infty}(x-\lambda)^r e^{-\lambda}\frac{\lambda^x}{x!} + \sum_{x=0}^{\infty}(x-\lambda)^r e^{-\lambda}\frac{x\lambda^{x-1}}{x!} \\
&= (-r)\sum_{x=0}^{\infty}(x-\lambda)^{r-1}e^{-\lambda}\frac{\lambda^x}{x!} + \sum_{x=0}^{\infty}(x-\lambda)^r e^{-\lambda}\frac{\lambda^x}{x!}\left[-1+\frac{x}{\lambda}\right] \\
&= (-r)\mu_{r-1} + \sum_{x=0}^{\infty}(x-\lambda)^r e^{-\lambda}\frac{\lambda^x}{x!}\frac{x-\lambda}{\lambda} \\
&= (-r)\mu_{r-1} + \frac{1}{\lambda}\sum_{x=0}^{\infty}(x-\lambda)^{r+1}e^{-\lambda}\frac{\lambda^x}{x!} \\
&= (-r)\mu_{r-1} + \frac{1}{\lambda}\mu_{r+1} \\
\Rightarrow \mu_{r+1} &= \lambda\left(\frac{\partial \omega_r}{\partial \lambda} + r\mu_{r-1}\right)
\end{aligned}$$

Corollary

$$\mu_0 = 1, \quad \mu_1 = 0.$$

1. $r=1, \quad \mu_2 = \lambda\left(\frac{\partial u_1}{\partial \lambda} + \mu_0\right) = \lambda = \text{variance}$
2. $r=2, \quad \mu_3 = \lambda\left(\frac{\partial \mu_2}{\partial x} + 2\mu_1\right) = \lambda(1+0) = \lambda$
3. $r=3, \quad \mu_4 = \lambda\left(\frac{\partial \mu_3}{\partial \lambda} + 3\mu_2\right) = \lambda(1+3\lambda) = 3\lambda^2 + \lambda$
4. Skewness $= \frac{\mu_1^3}{\mu_2^{3/2}} = \frac{\lambda}{\lambda^{3/2}} = \frac{1}{\sqrt{\lambda}} > 0$
   $\therefore$ positively skewed.
5. Kurtosis $= \frac{\mu_4}{\mu_2^2} - 3 = \frac{3\lambda^2+\lambda}{\lambda^2} - 3 = 3 + \frac{1}{\lambda} - 3 = \frac{1}{\lambda} > 0.$
   $\therefore$ leptokurtic.

7 Poisson Approximation of Binomial Distribution

Suppose, $X \sim \text{Bin}(n,p)$ and if the following conditions are satisfied.
1. $n \to \infty$. (The no of trails are infinitely large)
2. $p \to 0$ (probability of success in each trail is too small)
3. $\begin{cases} np \to \lambda & (\text{ finite } > 0) \\ np = \lambda & (\text{ finite } > 0) \end{cases}$

Example:

(1) $p = \frac{1}{n}, \Rightarrow np = 1, (n = \text{large }), \lambda = 1$

(2) $p = \frac{1}{n} + \frac{1}{n^2}, (n \text{ large }), n \to \infty, \Rightarrow p \to 0$

$$np = 1 + \frac{1}{n} \to 1, \lambda \to 1.$$

The distribution of X will converge in a Poisson Distribution with parameter λ.

Proof: $X \sim \text{Bin}(n, p)$.

$$p(x) = \binom{n}{x} p^x q^{n-x}$$

$$\begin{aligned}
\lim_{\substack{n\to\infty \\ p\to 0 \\ np=\lambda}} \binom{n}{x} p^x q^{n-x} &= \lim_{n\to\infty} \frac{n!}{x!(n-x)!} \left(\frac{\lambda}{n}\right)^x \left(1 - \frac{\lambda}{n}\right)^{n-x} \\
&= \frac{\lambda^x}{x!} \lim_{n\to\infty} \frac{n!}{(n-x)!} \frac{1}{n^x} \left(1 - \frac{\lambda}{n}\right)^{n-x} \\
&= \frac{\lambda^x}{x!} \lim_{n\to\infty} \left[\frac{n(n-1)(n-2)\dots(n-x+1)}{n^x}\right] \left(1 - \frac{\lambda}{n}\right)^{n-x} \\
&= \frac{\lambda^x}{x!} \lim_{n\to\infty} \underbrace{\left[\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\cdots\left(1 - \frac{x-1}{n}\right)\right]}_{} \frac{\left(1 - \frac{\lambda}{n}\right)^n\}}{\left(1 - \frac{\lambda}{n}\right)^x\}} \qquad (1)
\end{aligned}$$

Now, $\lim_{n\to\infty} \left[1 \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{x-1}{n}\right)\right] = 1$

$$\lim_{n\to\infty} \left(1 - \frac{\lambda}{n}\right)^n = e^{-\lambda}$$
$$\lim_{n\to\infty} \left(1 - \frac{\lambda}{n}\right)^x = e^{-\infty} = 0.$$

By (1), $\lim_{\substack{n\to\infty \\ p\to 0}} \binom{n}{x} p^x q^{n-x} = \frac{\lambda^x}{x!} e^{-\lambda}$ (PMF of Poisson distribution)

8 Poisson Approximation of Negative Binomial Distribution

Let $X \sim \text{NB}(r, p)$. Suppose the following conditions are satisfied,
1) $r \to \infty$
2) $q \to 0 \Leftrightarrow p \to 1$
3) $rq \to \lambda$ or $rq = \lambda$.

Then, the distribution of X will converge to a Poisson (λ) distribution.

Proof: $X \sim \text{NB}(r, p)$.

$$p(X = x) = \binom{x + r - 1}{x} p^r q^x$$

$$\begin{aligned}
\lim_{\substack{r\to\infty \\ q\to 0 \\ rq=1}} \binom{x + r - 1}{x} p^r q^x &= \lim_{r\to\infty} \frac{(x + r - 1)!}{x!(r-1)!} \left(1 - \frac{\lambda}{r}\right)^r \left(\frac{\lambda}{r}\right)^x \\
&= \frac{\lambda^x}{x!} \lim_{r\to\infty} \left[\frac{(x + r - 1)(x + r - 2)\dots(r+1)r}{ra}\right] \left(1 - \frac{\lambda}{r}\right)^r \\
&= \frac{\lambda^x}{x!} \lim_{r\to\infty} \left[\left(\frac{x-1}{r} + 1\right)\left(\frac{x-2}{r} + 1\right)\dots\left(1 + \frac{1}{r}\right)1\right] \left(1 - \frac{\lambda}{r}\right)^r
\end{aligned}$$

Now, $\lim_{r\to\infty} \left[\left(\frac{x-1}{r} + 1\right)\left(\frac{x-2}{r} + 1\right)\left(1 + \frac{1}{r}\right)1\right] = 1$
$\lim_{r\to\infty} (1 - \frac{\lambda}{r})^r = e^{-\lambda}$

$$\therefore \lim_{\substack{r\to\infty \\ q\to 0 \\ rq=\lambda}} \binom{x+r-1}{x} p^r q^x = e^{-\lambda}\frac{\lambda^x}{x_0^!} \text{(PMF of Poisson Distribution)}$$

Remark

If $X \sim \text{Poi}(\lambda)$ for any function $g(x)$,

$$\begin{aligned} E(Xg(x)) &= \sum_{x=0}^{\infty} xg(x)e^{-\lambda}\frac{\lambda^x}{x!} \\ &= \sum_{x=1}^{\infty} g(x)e^{-\lambda}\frac{\lambda^x}{(x-1)!} \\ &= \sum_{y=x-1=0}^{\infty} g(y+1)e^{-\lambda}\frac{\lambda^{y+1}}{y!} \\ &= \lambda\sum_{y=0}^{\infty} g(y+1)e^{-\lambda}\frac{\lambda^y}{y!} \\ &= \lambda E[g(Y+1)] \equiv \lambda E[g(x+1)] \end{aligned}$$

Now,

1. Let, $g(x) = c(\neq 0)$
   E(Xc) $=\lambda$ E(c)$=\lambda$ c $\Rightarrow$ E(x)$=\lambda$
2. Let, $g(x) = x$,
   $E\left(X^2\right) = \lambda E(X+1) = \lambda(\lambda+1) = \lambda^2+\lambda$
   $V(X) = E\left(X^2\right) - E^2(X) = \lambda^2+\lambda-\lambda^2 = \lambda$

9 Properties

1. If $X_i \sim \text{Poi}\left(\lambda_i\right), i = 1, 2$ independently, show that $X_1 + X_2 \sim \text{Poi}\left(\lambda_1+\lambda_2\right)$ (Reproductive Property).

   Proof:

$$\begin{aligned} Y &= X_1 + X_2 \\ P(Y=y) = P\left(X_1+X_2=y\right) &= P\left(X_1=x, X_2=y-x\right) \\ &= \sum_{x=0}^{\infty} P\left(X_1=x\right)P\left(X_2=y-x\right) \\ &= \sum_{x=0}^{\infty} e^{-\lambda_1}\frac{\lambda_1^x}{x!}e^{-\lambda_2}\frac{\lambda_2^{y-x}}{(y-x)!} \\ &= \frac{e^{-(\lambda_1+\lambda_2)}}{y!}\sum_{x=0}^{\infty}\lambda_1^x\lambda_2^{y-x}\frac{y!}{x!(y-x)!} \\ &= \frac{e^{-(\lambda_1+\lambda_2)}}{y!}\left(\lambda_1+\lambda_2\right)^y \\ \therefore Y = X_1+X_2 &\sim \text{Poi}\left(\lambda_1+\lambda_2\right) \end{aligned}$$

2. Find the conditional distribution of $(X_1 \mid X_1 + X_2)$.

$$
\begin{aligned}
P\left[X_1 = x \mid X_1 + X_2 = y\right] &= \frac{P\left(X_1 = x, X_1 + X_2 = y\right)}{P\left(X_1 + X_2 = y\right)} \\
&= \frac{P\left(X_1 = x\right) P\left(X_2 = y - x\right)}{P\left(X_1 + X_2 = y\right)} \\
&= \frac{e^{-\lambda_1}\lambda_1^x/x! \quad e^{-\lambda_2}\lambda_2^{y-x}/(y-x)!}{e^{-(\lambda_1+\lambda_2)}\left(\lambda_1+\lambda_2\right)^y/y!} \\
&= \frac{y!}{x!(y-x)!}\left(\frac{\lambda_1}{\lambda_1+\lambda_2}\right)^x\left(1-\frac{\lambda_1}{\lambda_1+\lambda_2}\right)^{y-x}
\end{aligned}
$$

So, $(X_1 \mid X_1 + X_2 = y) \sim \text{Bin}\left(y, \dfrac{\lambda_1}{\lambda_1+\lambda_2}\right)$

3. $X \sim \text{Poi}(\lambda)$, show that, a) $P\left(X \leqslant \dfrac{\lambda}{2}\right) < \dfrac{4}{\lambda}$, b) $P(X > 2\lambda) < \dfrac{1}{\lambda}$

a)

$$
\begin{aligned}
P\left(X \leqslant \frac{\lambda}{2}\right) = P\left(-X \geqslant -\frac{\lambda}{2}\right) &= P\left(\lambda - X \geqslant \frac{\lambda}{2}\right) \\
&= P\left((\lambda - X)^2 \geqslant \frac{\lambda^2}{4}\right) \\
&\leqslant \frac{E(\lambda - X)^2}{\lambda^2/4} = \frac{4\lambda}{\lambda^2} = \frac{4}{\lambda}
\end{aligned}
$$

b)

$$
\begin{aligned}
P(X > 2\lambda) &= P(X - \lambda > \lambda) \\
&= P\left((X-\lambda)^2 > \lambda^2\right) \\
&\leqslant \frac{E(X-\lambda)^2}{\lambda^2} \\
&= \frac{\lambda}{\lambda^2} = \frac{1}{\lambda}
\end{aligned}
$$

10 Cumulative Distribution

If $X \sim \text{Poi}(\lambda)$, for any non-negative integer k, $P(X \leqslant k) = \dfrac{1}{\Gamma(k+1)}\int_\lambda^\infty e^{-k}x^k dx$.

Proof:

$$
\begin{aligned}
P(X \leqslant k) &= \sum_{x=0}^{k} e^{-\lambda}\frac{\lambda^x}{x!} = g(\lambda), (\text{say}) \rightarrow \text{ a continuous function of } \lambda \\
g'(\lambda) &= -\sum_{x=0}^{k} e^{-\lambda}\frac{\lambda^x}{x!} + \sum_{x=0}^{k} e^{-\lambda}\frac{x\lambda^{x-1}}{x!} \\
&= \sum_{x=0}^{k}\left\{\frac{\lambda^{x-1}}{(x-1)!}e^{-\lambda} - e^{-\lambda}\frac{\lambda^x}{x!}\right\} \\
&= e^{-\lambda}\left(-\frac{\lambda^0}{0!} + \frac{\lambda^0}{0!} - \frac{\lambda^1}{1!} + \frac{\lambda^1}{1!} - \frac{\lambda^{(k-1)}}{(k-1)!} + \ldots - \frac{\lambda^k}{k!}\right) \\
&= -e^{-\lambda}\frac{\lambda^k}{k!}
\end{aligned}
$$

We can say, $g'(\mu) = -e^{-\mu}\frac{\mu^k}{k!}$

$$\int_\lambda^\infty g'(\mu)d\mu = -\int_\lambda^\infty e^{-\mu}\frac{\mu^k}{k!}d\mu$$

$$\Rightarrow g(\infty) - g(\lambda) = -\frac{1}{\Gamma(k+1)}\int_\lambda^\infty e^{-\mu}\mu^k d\mu$$

$$g(\lambda) = \frac{1}{\Gamma(k+1)}\int_\lambda^\infty e^{-\mu}\mu^k d\mu$$

Now, $P(X \leqslant k) = \sum_{x=0}^{k} f_X(x) = g(\lambda)$, $g(\lambda)$ is a decreasing function of λ.

The tail function of X is given by,

$$\begin{aligned}
P(X > x) &= 1 - P(X \leqslant x) \\
&= 1 - \frac{1}{\Gamma(k+1)}\int_\lambda^\infty e^{-\mu}\mu^k dw \\
&= \frac{1}{\Gamma(k+1)}\int_0^\infty e^{-\mu}\mu^k d\mu - \frac{1}{\Gamma(k+1)}\int_\lambda^\infty e^{-\mu}\mu^k d\mu \\
&= \frac{1}{\Gamma(k+1)}\int_0^\lambda e^{-\mu}\mu^k d\mu \\
P(X \leqslant k) &= \frac{1}{\Gamma(k+1)}\int_\lambda^\infty e^{-\mu}\mu^{\overline{k+1}-1} d\mu \\
P(X > k) &= \frac{1}{\Gamma(k+1)}\int_0^\lambda e^{-\mu}\mu^{\overline{k+1}-1} d\mu \\
&= \Gamma(\lambda^{(k+1)}) \text{ (Just a notation for incomplete gamma function)}
\end{aligned}$$

4. If $X \sim \text{Poi}(\lambda)$ for any non-negative k, show that $P(X \geqslant k) < \frac{\lambda^k}{k!}$.

$$\begin{aligned}
P(X \geqslant k) = P(X > k-1) &= \frac{1}{\Gamma k}\int_0^\lambda e^{-u}u^{k-1}du \\
&< \frac{1}{\Gamma k}\int_0^\lambda u^{k-1}du \quad \left(\because e^{-u} > 0\right) \\
&= \frac{1}{\Gamma k}\left[\frac{u^k}{k}\right]_0^\lambda \\
&= \frac{1}{k(k-1)!}\lambda^k = \frac{\lambda^k}{k!}
\end{aligned}$$

5. The number of bacteria in a source of a liquid is known to be a Poisson random variable with mean λ per *cc*. If n one cc tubes are filled with liquid, what is the probability distribution of the number of test tubes that show growth? (ie. have at least one bacteria)

X = No of test tubes that shows growth.
Y = No of bacteria in *cc*.
Given, $Y \sim \text{Poi}(\lambda)$,

$$\begin{aligned}
P(\text{at least one bacteria}) &= 1 - P(\text{ no bacteria }) \\
&= 1 - e^{-\lambda}\frac{\lambda^0}{0!} = 1 - e^{-\lambda}
\end{aligned}$$

P (one test tube will show a growth)

$$\begin{aligned}
&= P(Y \geqslant 1) \\
&= 1 - P(Y = 0) = 1 - e^{-\lambda}\frac{\lambda^0}{0!} = 1 - e^{-\lambda}
\end{aligned}$$

So, the distribution of X is, $X \sim \text{Bin}(n, p)$ where $p = 1 - e^{-\lambda}$

$$\begin{aligned} P(X = x) &= \binom{n}{x} p^x q^{n-x} \\ &= \binom{n}{x} \left(1 - e^{-\lambda}\right)^x \left(e^{-\lambda}\right)^{n-x} \end{aligned}$$

6. If $X \sim \text{Bin}(n, p)$ and $np = \lambda$, show that

$$\frac{\lambda^k}{k!}\left(1 - \frac{\lambda}{n}\right)^{n-k} \geqslant P(X = k) \geqslant \frac{\lambda^k}{k!}\left(1 - \frac{k}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k}$$

Proof:

$$\begin{aligned} P(X = k) &= \binom{n}{k} p^k (1-p)^{n-k} \\ &= \frac{n(n-1)\dots(n-k+1)}{k!}\left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} \\ &= \frac{\frac{n}{n}\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\dots\left(1 - \frac{k-1}{n}\right)}{k!} \lambda^k \left(1 - \frac{\lambda}{n}\right)^{n-k} \end{aligned}$$

Now, $\left(1 - \frac{1}{n}\right) < 1, \left(1 - \frac{2}{n}\right) < 1, \dots, \left(1 - \frac{k-1}{n}\right) < 1$

$$\Rightarrow \left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\dots\left(1 - \frac{k-1}{n}\right) < 1$$

$$P(X = k) \leqslant \frac{\lambda^k}{k!}\left(1 - \frac{\lambda}{n}\right)^{n-k}$$

Now $1 - \frac{1}{n} > 1 - \frac{k}{n}, \quad 1 - \frac{2}{n} > 1 - \frac{k}{n}, \dots, 1 - \frac{k-1}{n} > 1 - \frac{k}{n},$

$$\Rightarrow \left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\cdots\left(1 - \frac{k-1}{n}\right) > \left(1 - \frac{k}{n}\right)^k$$

$$\therefore P(X = k) \geqslant \frac{\lambda^k}{k!}\left(1 - \frac{\lambda}{n}\right)^{n-k}\left(1 - \frac{k}{n}\right)^k$$

Taking limits on both side,

$$\Rightarrow e^{-\lambda}\frac{\lambda^k}{k!} \geqslant \lim_{n\to\infty} P(X = k) \geqslant e^{-\lambda}\frac{\lambda^k}{k!}$$

By Sandwich theorm, we can say, $\lim_{n\to\infty} P(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}$

11 Definition of Poisson Process

Suppose, we are observing the occurrence of an event over time or space or length. Assuming that there is a quantity such that,

1. The probability that exactly one happening will occur in a small time interval of length t approximately.
2. The probability that more than one happening will occur in a small time interval of length this negligible.
3. The number of happenings in non-overlapping intervals are independent.

Under the above condition, it can be shown that the number of occurrence in time interval of length t has a distribution.

Also, the number of occurrence in time interval of length "kt" has a distribution. Here, is the mean rate of occurrence of the event in a unit time interval.

It will follow Poi ($\lambda = \mu kt$) distribution.

12 Incomplete Gamma and Beta Function

Consider standard Gamma$(n, \theta = 1)$ distribution with $f(x) = \begin{cases} \frac{e^{-x}x^{n-1}}{\Gamma n}, 0 < x < 1 \end{cases}$

For, $0 < x < \infty$, $\quad F(x) = \int_0^x \frac{e^{-t}t^{n-1}}{\Gamma n}dt = \frac{\int_0^x e^{-t}t^{n-1}dt}{\int_0^\infty e^{-t}t^{n-1}dt} = \Gamma(x^{(n)})$

Tabulated values are obtained from Karl-Pearason Table.

Use: If $X \sim \text{Poi}(\lambda), P(X \geqslant k) = \sum_{x=k}^{\infty} e^{-\lambda}\frac{\lambda^x}{x!} = \int_0^\lambda \frac{e^{-t}t^{k-1}}{\Gamma k}dt$

Conversely, if X is a discrete random variable with DF $F(k) = \int_\lambda^\infty \frac{e^{-t}t^{k-1}}{\sqrt{k}}dt, k = 0, 1, 2, \ldots$, show that $X \sim \text{Poi}(\lambda)$.

Relationship between Poisson and Gamma Distribution

Let, the no of occurrence of an event over time has a Poisson distribution. Let t_n be the waiting time for the nth occurrence of the event, then t_n has a Gamma Distribution.

13 Examples of Poisson Distribution

1. Under the condition of Poisson process, some random variables involving counts of happenings of an event over time, space or length, can be modeled realistically, by a Poisson distribution. Following are such examples of Poisson distribution.
   a) The number of telephone calls in an hour in a large business house.
   b) The number of defects in an unit of a material.
   c) The number of natural deaths in a year in a given region.

2. Each of the following and numerous others random variables are approximately following Poisson distribution because of Poisson approximation to Binomial.
   a) The number of misprints in a page of a book.

Here, we can assume that there is a small probability p that each letter typed will be misprinted and the number of letters n in a page is large. Hence, the number of misprints in a page will approximately follow a Poisson distribution. $(np = \lambda)$

If the printing quality is poor or bad, the probability p that each letter typed will be misprinted, is not a small quantity, then the Poisson approximation to Binomial is not applicable.

b) The number of individuals in a large community living upto 100 years of age.